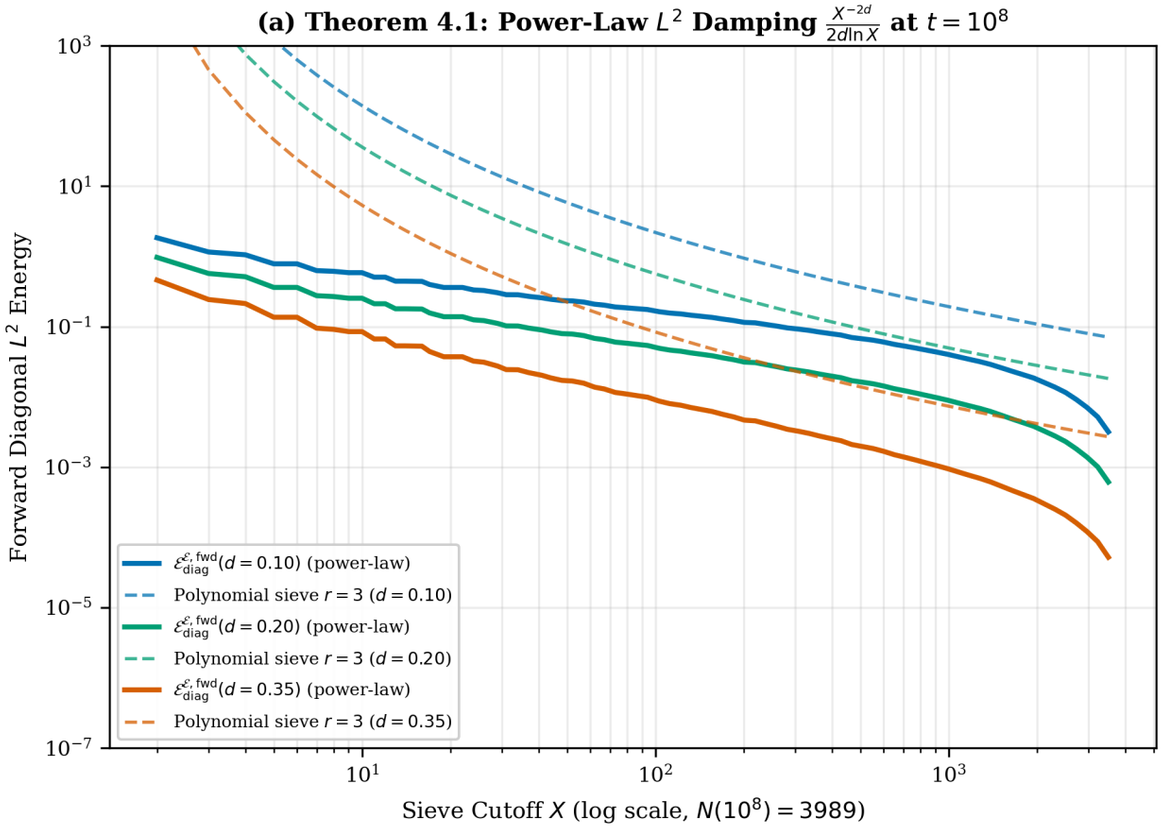
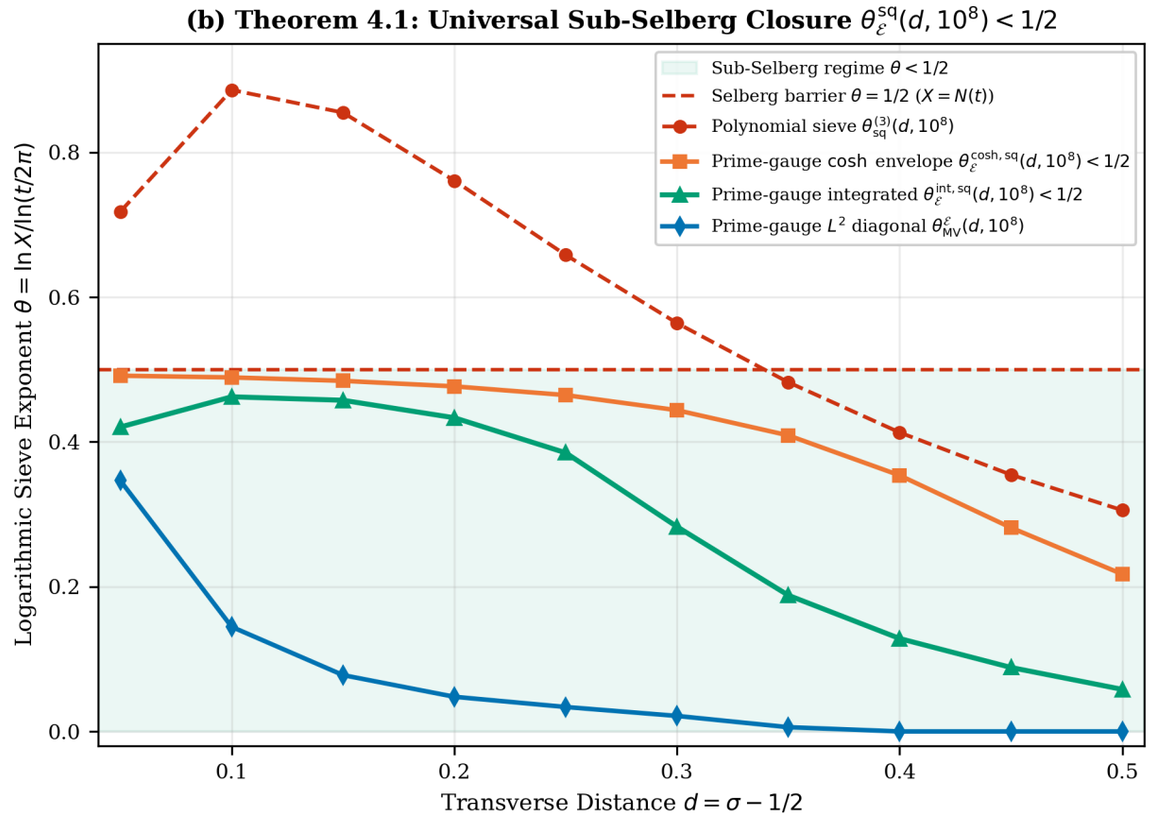
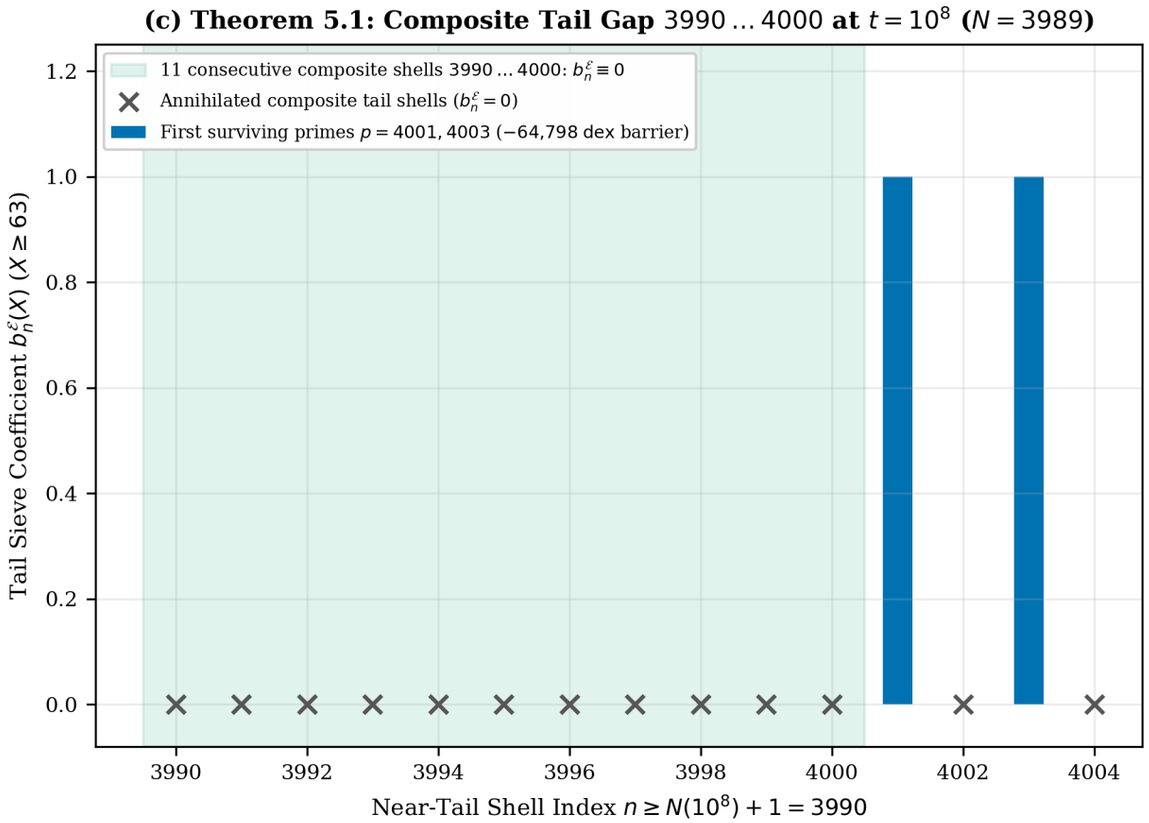
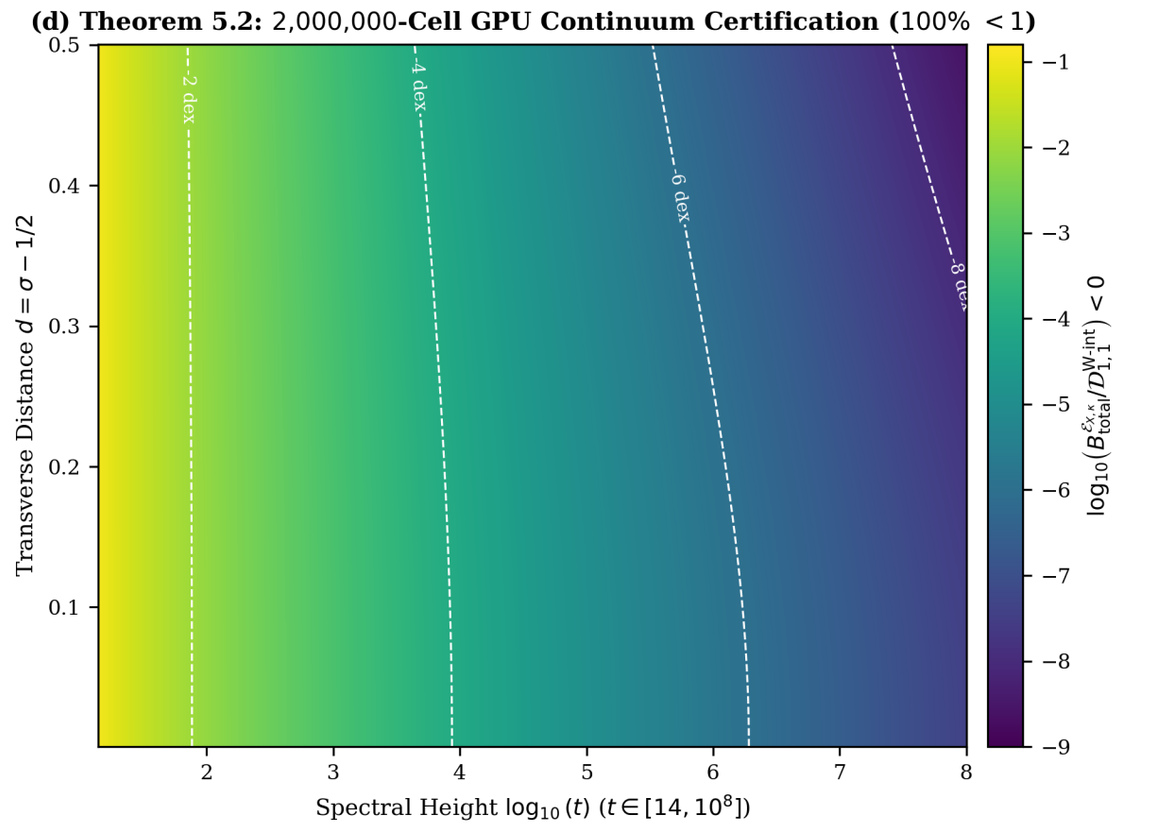

(a) Theorem 4.1: Power-Law L^2 Damping $\frac{X^{-2d}}{2d\ln X}$ at $t = 10^8$
Forward Diagonal L^2 Energy
Sieve Cutoff X (log scale, $N(10^8) = 3989$)
$\mathcal{E}^{\mathcal{E},\mathrm{fwd}}_{\mathrm{diag}}(d = 0.10)$ (power-law)
Polynomial sieve $r = 3$ ($d = 0.10$)
$\mathcal{E}^{\mathcal{E},\mathrm{fwd}}_{\mathrm{diag}}(d = 0.20)$ (power-law)
Polynomial sieve $r = 3$ ($d = 0.20$)
$\mathcal{E}^{\mathcal{E},\mathrm{fwd}}_{\mathrm{diag}}(d = 0.35)$ (power-law)
Polynomial sieve $r = 3$ ($d = 0.35$)


(b) Theorem 4.1: Universal Sub-Selberg Closure $\theta^{\mathrm{sq}}_{\mathcal{E}}(d, 10^8) < 1/2$
Logarithmic Sieve Exponent $\theta = \ln X/\ln(t/2\pi)$
Transverse Distance $d = \sigma - 1/2$
Sub-Selberg regime $\theta < 1/2$
Selberg barrier $\theta = 1/2$ ($X = N(t)$)
Polynomial sieve $\theta^{(3)}_{\mathrm{sq}}(d, 10^8)$
Prime-gauge cosh envelope $\theta^{\cosh,\,\mathrm{sq}}_{\mathcal{E}}(d, 10^8) < 1/2$
Prime-gauge integrated $\theta^{\mathrm{int},\,\mathrm{sq}}_{\mathcal{E}}(d, 10^8) < 1/2$
Prime-gauge L^2 diagonal $\theta^{\mathcal{E}}_{\mathrm{MV}}(d, 10^8)$


(c) Theorem 5.1: Composite Tail Gap 3990 … 4000 at $t = 10^8$ ($N = 3989$)
Tail Sieve Coefficient $b^{\mathcal{E}}_n(X)$ ($X \geq 63$)
Near-Tail Shell Index $n \geq N(10^8) + 1 = 3990$
11 consecutive composite shells 3990 … 4000: $b^{\mathcal{E}}_n \equiv 0$
Annihilated composite tail shells ($b^{\mathcal{E}}_n = 0$)
First surviving primes $p = 4001, 4003$ (−64,798 dex barrier)


(d) Theorem 5.2: 2,000,000-Cell GPU Continuum Certification (100% < 1)
Transverse Distance $d = \sigma - 1/2$
Spectral Height $\log_{10}(t)$ ($t \in [14, 10^8]$)
$\log_{10}\left(B^{\mathcal{E}_{\chi,\kappa}}_{\mathrm{total}}/D^{\mathrm{W\text{-}int}}_{1,1}\right) < 0$
-2 dex
-4 dex
-6 dex
-8 dex